Nikhil Tripathi

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EDUCATION

University of California, Los Angeles (UCLA) — Los Angeles, CA
B.S. in Computer Science | GPA: 3.65 — *Expected June 2028*
Coursework: Data Structures, Algorithms & Complexity, Object-Oriented Programming, Operating Systems, Computer Organization, Software Construction, Programming Languages, Artificial Intelligence, Logic Design, Computing Theory, Discrete Mathematics, Linear Algebra, Differential Equations, Probability

EXPERIENCE

Visa | Global Tokenization Services — Foster City, CA
Software Engineer Intern — *Jun 2026 - Aug 2026*
- Built a GenAI conversational assistant using Spring Boot, Spring AI, and Redis to query tokenized payment data across 3 live financial sources, cutting root-cause transaction investigation time from 24+ hours to near-instant.
- Engineered a hybrid search pipeline combining pgvector semantic retrieval with exact SQL aggregation over PostgreSQL event logs, eliminating LLM sampling errors and guaranteeing exact metric calculations across full event histories.
- Unified transaction reporting across 750+ QA tokens and built a 64-event test dataset covering edge-case failure modes and real Visa error codes to enforce cross-source grounding and deterministic query routing.

UCLA Stress Lab | Semel Institute for Neuroscience and Health — Los Angeles, CA
Undergraduate ML Researcher — *Mar 2026 - Present*
- Building and maintaining an offline Python audio processing pipeline combining Librosa acoustic feature extraction and local Whisper ASR transcription to accurately model voice stress and clinical sentiment outcomes.
- Automating comprehensive feature extraction and audio chunking workflows across 1,400+ clinical audio recordings, enabling robust Pandas-driven sentiment analysis and trend visualization for ongoing clinical research.
- Built automated transcript quality control, noise exclusion heuristics, and regression-to-mean exploratory analysis pipelines.

Rivet Industries — Palo Alto, CA
Software Engineer Intern — *Jun 2025 - Sep 2025*
- Engineered scalable Python ETL pipelines processing 100,000+ SPAD sensor images to optimize complex camera hardware configurations, increasing image signal processing performance by 15%.
- Reduced manual testing duration by 40% through automated test suites integrated into Docker CI/CD pipelines.
- Deployed Linux containerized internal infrastructure with 99.9% operational uptime and role-based access security controls.

PROJECTS

CampusTrendsML | *Python, Sentence-Transformers, BERTopic, XGBoost, scikit-learn*
- Engineered a robust offline ML pipeline utilizing Sentence-Transformers and BERTopic to cluster and track emerging topics across 10,000+ daily student discussion posts.
- Trained a high-performance XGBoost classifier leveraging TF-IDF vectorization, sentiment polarity, and temporal metadata to accurately predict dynamic post engagement trends.

UCLA Applicant Tracking System | *React, Node.js, PostgreSQL, Docker*
- Developed a comprehensive full-stack club recruitment portal featuring custom form builders, candidate evaluation metrics, and advanced review analytics validated across 500+ mock applications.
- Successfully reduced deployment overhead using Dockerized React and Node.js environments with automated PostgreSQL schema migrations and secure, sanitized four-state authentication flows.

PropShop | *Python, Asyncio, Playwright, Selenium, Express.js, React*
- Developed asynchronous Python web scrapers using Playwright and Asyncio to monitor real-time odds variations across multiple betting platforms, feeding a robust Express.js API and React dashboard.
- Automated arbitrage detection and order execution workflows, generating $5,000+ in net profit over 5 months.

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, TypeScript, SQL (PostgreSQL, MySQL), Bash/Shell, HTML/CSS, C
Frameworks & Libraries: Spring Boot, Spring AI, React, Node.js, Express.js, PyTorch, XGBoost, Transformers, BERTopic, Pandas, NumPy, scikit-learn, Librosa, Keras
Tools, Databases & Platforms: Docker, Linux, Git, GitHub Actions, CI/CD, PostgreSQL, pgvector, Redis, REST APIs, Microservices, Kubernetes, Maven, Flyway, Bruno, Postman